



ESP32-S3 4.2inch RLCD Development Board, 300×400, Supports Wi-Fi &BLE Dual-mode Communication And AI Voice Interaction

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Color: Without Batt



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★★★★★
Color:Without Batt

Very satisfied, as always with this store :) item as described, exceptionally well packed, no issues

S***a | 14 Feb 2026

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Specifications

High-concerned chemical	None	Type	Module
Origin	Mainland China	Condition	New

Description

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ESP32-S3 4.2inch RLCD development board

E-paper-like Reading Experience, Faster Refresh Response

AI Voice Interaction, Temperature & Humidity Monitoring, DIY Learning & Development



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ESP32-S3



2.4 GHz Wi-Fi



BLE 5



SHTC3



Black / White



Reflective LCD



4.2"

300 × 400
Resolution

Audio Capture

PCF85063
RTC Chip

TF Card Slot



Audio Output

Programmable
Button

16MB Flash



8MB PSRAM

DIY Expansion
Interface

Version Options

**ESP32-S3-RLCD-4.2-EN**

Without 18650 Lithium Batt

Features

The **ESP32-S3-RLCD-4.2** is a RLCD AIoT development board based on ESP32-S3, supporting Wi-Fi and BLE dual-mode communication. Onboard 4.2inch RLCD with low power consumption, e-paper-like display experience, and faster refresh response. Integrated with audio codec circuit, dual microphones, speaker, SHTC3 high-precision temperature and humidity sensor, TF card slot, RTC header, and Lithium Batt charging and discharging management circuit, etc. Reserved USB, UART, I2C, and multiple GPIO interfaces for easy expansion. Supports AI Voice Interaction, temperature and humidity monitoring, and IoT control, making it suitable for DIY desktop smart ornaments, electronic calendars, AI smart devices, and other applications. Also ideal for product prototyping.

- Adopts high-performance Xtensa 32-bit LX7 dual-core processor, up to 240MHz main frequency
- Supports 2.4GHz Wi-Fi and Bluetooth 5 (LE), with onboard antenna
- Built-in 512KB Static RAM, 384KB ROM, with integrated 16MB Flash and 8MB PSRAM
- Onboard 4.2inch RLCD, 300 × 400 resolution, features reflective imaging and no backlight required
- Equipped with a dual-microphone array, enabling audio algorithms such as noise reduction and echo cancellation, suitable for accurate voice recognition and near-field / far-field voice wake-up applications
- Onboard PCF85063 RTC chip and SHTC3 temperature & humidity sensor for accurate RTC management and environmental monitoring
- Onboard 18650 Lithium Batt holder and RTC backup Batt header. The RTC backup Batt header supports a rechargeable RTC Batt and dual modes of main Batt power supply and RTC independent power supply
- Onboard TF card slot for external storage of images or files
- Onboard programmable KEY and BOOT side buttons for customized function development
- Reserved 2 × 8PIN 2.54mm pitch female header for convenient external expansion

Specifications

display panel	RLCD	Display size	4.2inch
resolution	300 × 400	grey scale	2
interface	SPI	Display Mode	reflective
Driver IC	ST7305	display color	black, white

Onboard Audio Codec Chip

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Supports AI Voice Interaction

Allows access to online large model platforms such as DeepSeek, Doubao, etc

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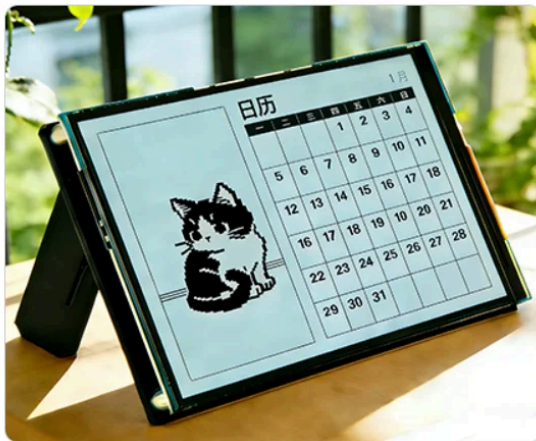
Reflective Liquid Crystal Display

The **Reflective Liquid Crystal Display (RLCD)** is a specialized type of LCD that efficiently utilizes ambient light (such as natural sunlight or indoor illumination) instead of a backlight, achieving both energy efficiency and environmental friendliness. Thanks to its reflective design, RLCDs perform exceptionally well in bright environments, where conventional LCDs may suffer from glare and reduced visibility. By effectively reflecting external light, the RLCD can provide a clear, high-quality visual experience.

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Application Scenarios

Ideal for DIY desktop smart ornaments, electronic calendars, AI smart devices, outdoor displays and more



Supports ESP-IDF, Arduino IDE

Comprehensive SDK, dev resources, and tutorials to help you easily get started



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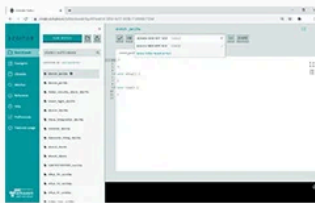
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ESP-IDF

ESP-IDF

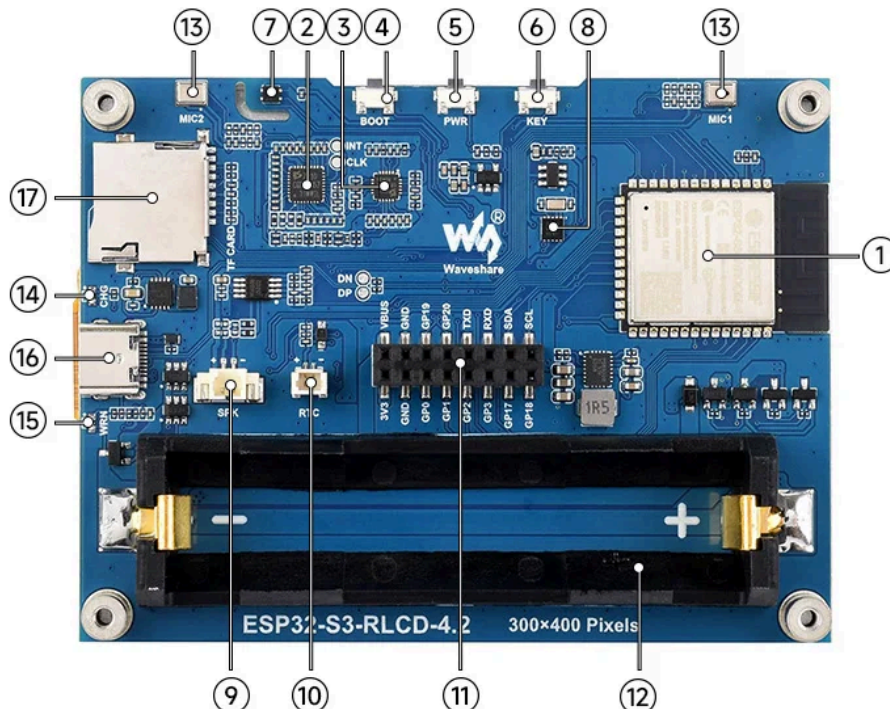
With free open source development tools, supports IDEs such as VSCode and Eclipse easier for developers to use.



Arduino IDE

Arduino IDE is an open source electronic prototyping platform, convenient and flexible, easy to get started.

What's On Board



- ADC chip for echo cancellation circuit

3. **ES8311**
low-power audio codec chip

4. **BOOT button**
Press it when powering on to enter download mode

5. **PWR button**
long press to power off, short press to power on

6. **KEY button**
customizable for other functions

7. **SHTC3 temperature and humidity sensor**
provides ambient temperature and humidity measurement, enabling environmental monitoring function

8. **PCF85063**
RTC chip with timekeeping function
1. **MX1.25 2PIN speaker header**
audio output signal, for connecting external speaker

2. **RTC independent power supply header**
supports PH1.0 rechargeable RTC Batt only

3. **2 × 8PIN 2.54mm pitch female header**

4. **18650 Lithium Batt holder**

5. **Dual-microphone array design**
dual-microphone array, paired with ES7210 to achieve echo cancellation

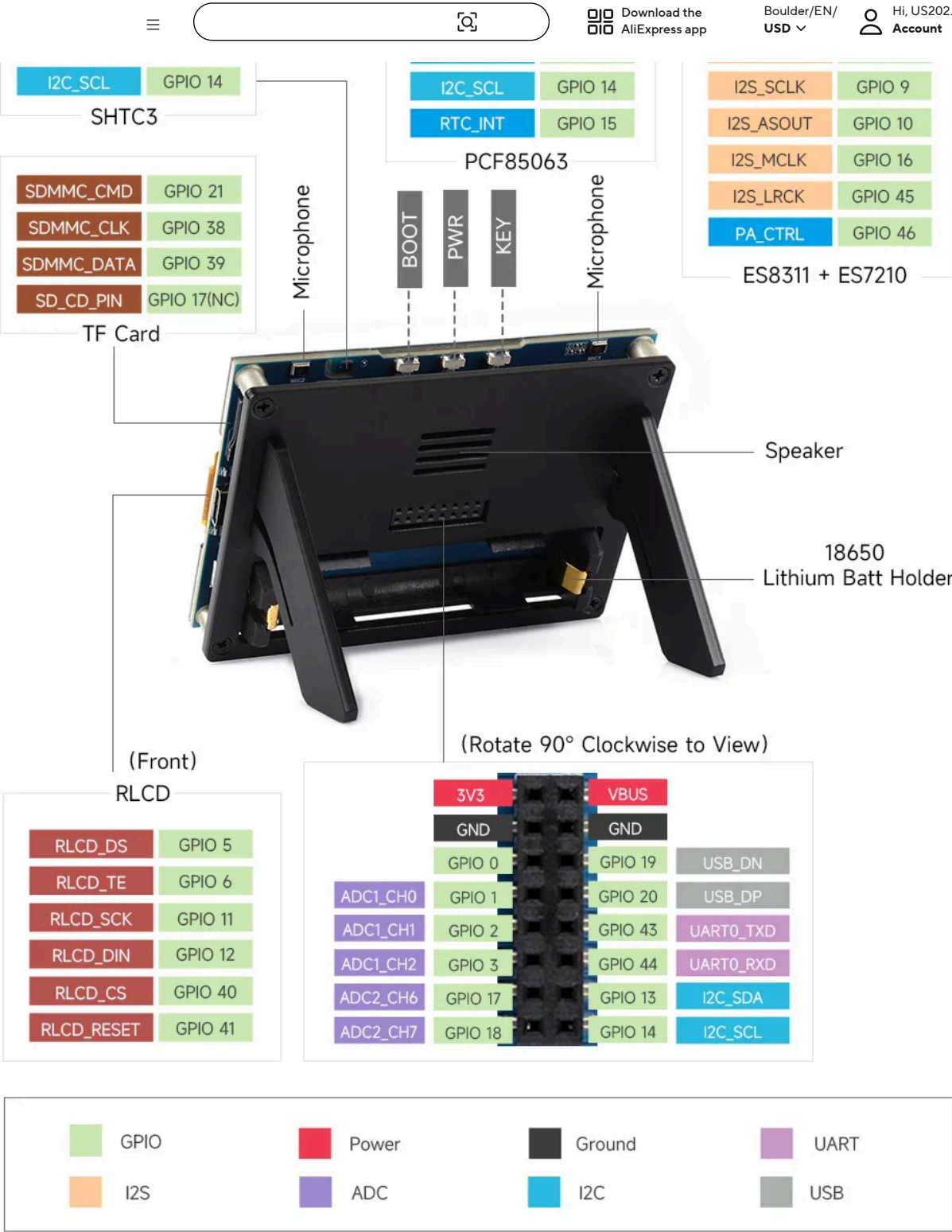
6. **CHG charging indicator**
off when fully charged

7. **WRN warning indicator**
stays on when Batt is reverse-connected

8. **Type-C port**
for program burning and log printing

9. **TF card slot**
supports FAT32 TF card for data expansion

Pin Definition



I2S_SCLK

GPIO 9

I2S_ASOUT

GPIO 10

I2S_MCLK

GPIO 16

I2S_LRCK

GPIO 45

PA_CTRL

GPIO 46

ES8311 + ES7210

Microphone

BOOT

PWR

KEY

Microphone

Speaker

18650 Lithium Batt Holder

RLCD_DS

GPIO 5

RLCD_TE

GPIO 6

RLCD_SCK

GPIO 11

RLCD_DIN

GPIO 12

RLCD_CS

GPIO 40

RLCD_RESET

GPIO 41

RLCD

3V3

GND

GPIO 0

GPIO 1

GPIO 2

GPIO 3

GPIO 17

GPIO 18

VBUS

GND

GPIO 19

GPIO 20

GPIO 43

GPIO 44

GPIO 13

GPIO 14

ADC1_CH0

ADC1_CH1

ADC1_CH2

ADC2_CH6

ADC2_CH7

USB_DN

USB_DP

UART0_TXD

UART0_RXD

I2C_SDA

I2C_SCL

(Front)

(Rotate 90° Clockwise to View)

GPIO

I2S

Power

ADC

Ground

I2C

UART

USB

Outline Dimensions

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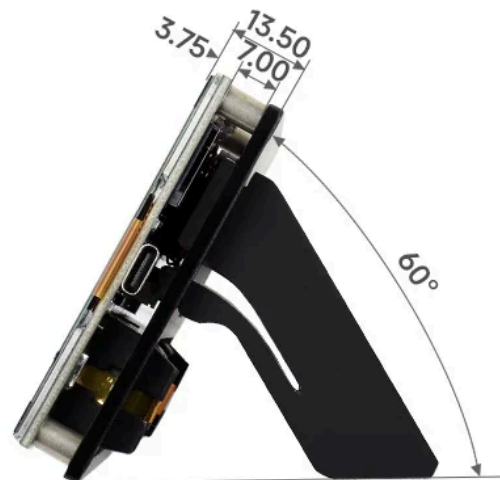
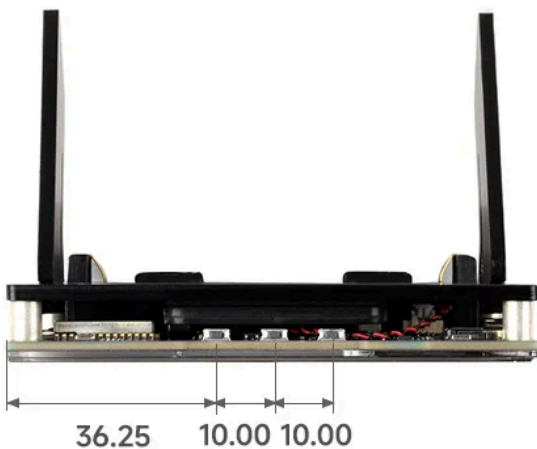
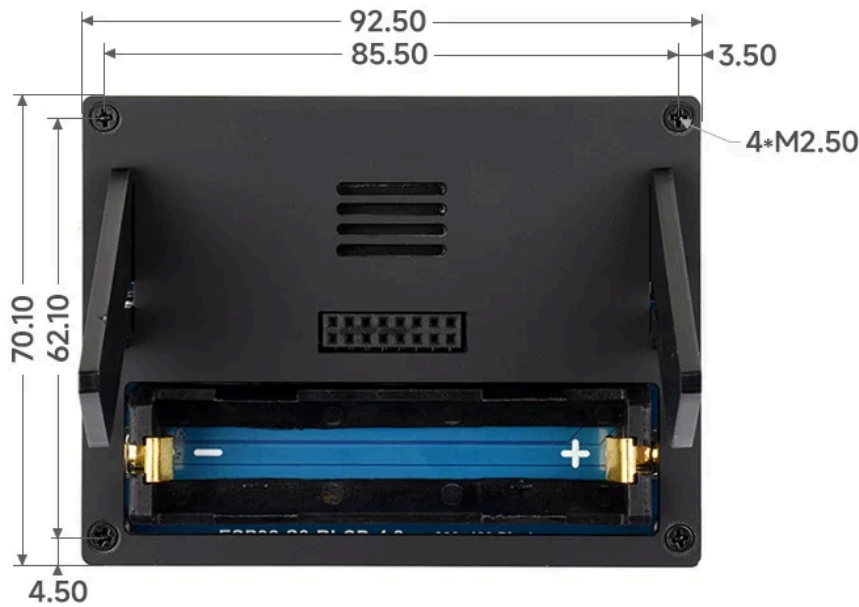
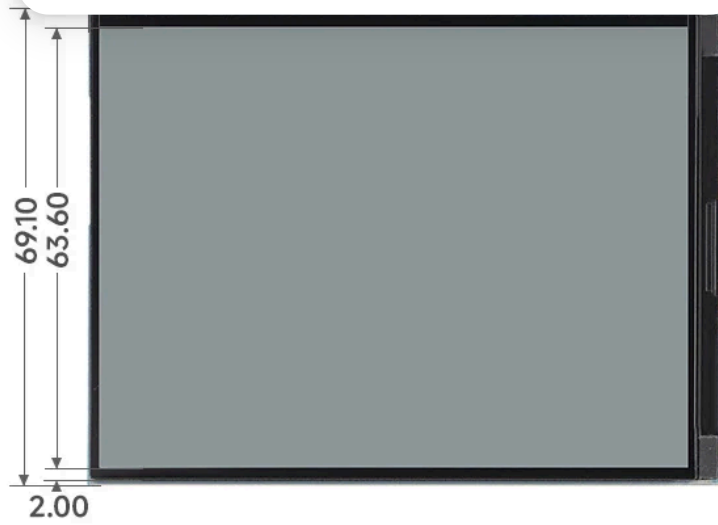
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